

Skills Summary

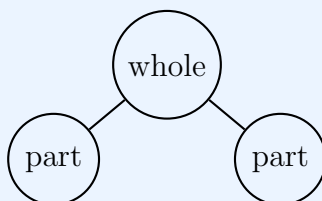
Number Bonds to 10 — Fixing an Incomplete or Duplicated Family

Use this reference while completing your worksheet or when studying.

Skill 1 — Finding the missing part of a bond

The model:

A number bond holds the **whole** on top and its two **parts** below.



part + part = whole **whole – part = the other part**

A part is always *smaller* than the whole. To find a missing part, take the part you know away from the whole — never add.

Example: The whole is 7 and one part is 2. Find the other part.

Step 1. A part is missing and the whole is known, so this is a take-away: whole minus the known part.

Step 2. Take 2 away from 7: two steps back from 7 lands on 5. Check by adding the parts back: $2 + 5 = 7$ ✓ The missing part is **5**.

Try it: The whole is 8 and one part is 6. Find the other part.

check: 2

Skill 2 — Spotting a fact written twice

How to check a family for a repeat:

- Count the **adding** facts. There must be exactly **two**.
- Count the **take-away** facts. There must be exactly **two**.
- Each take-away fact starts with the whole and removes a **different** part. If both remove the *same* part, one is a repeat.

Cross out the repeat, then write the fact that takes away the *other* part.

Example: A family for parts 3 and 7: $3 + 7 = 10$, $7 + 3 = 10$, $10 - 7 = 3$, $10 - 7 = 3$.

Step 1. Both take-away facts remove the same part 7, so the second one is a repeat and the fact that removes the part 3 is missing.

Step 2. Remove the other part instead: $10 - 3$ leaves the part 7, and $3 + 7 = 10$ checks it ✓ The missing fact is $10 - 3 = 7$.

Try it: A family for parts 2 and 6: $2 + 6 = 8$, $6 + 2 = 8$, $8 - 6 = 2$, $8 - 2 = 6$. Write the missing take-away fact $8 - 2 = ?$

check: 9

Skill 3 — Writing the whole family of four facts

Rule — every bond makes four facts:

- part + other part = whole
- other part + part = whole
- whole – part = other part
- whole – other part = part

Two adding facts, two take-away facts. If you have written fewer than four, the family is **incomplete**.

Example: Write the whole family for the parts 4 and 5.

Step 1. The two parts are known and the whole is not, so add the parts first — the whole is what the family is built on.

Step 2. $4 + 5 = 9$, so the family is $4 + 5 = 9$, $5 + 4 = 9$, $9 - 4 = 5$, $9 - 5 = 4$. The whole is 9.

Try it: Find the whole for the parts 2 and 5, then write all four facts.

check: 7 = whole

(!) Watch out: Turning an adding fact around ($5 + 4$ instead of $4 + 5$) makes a *new* fact, but turning a take-away fact around does not — $10 - 3$ and $10 - 7$ are different facts, while writing $10 - 7$ twice is the same fact twice.